Louisa Swintosky

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PROFESSIONAL SUMMARY

Detail-oriented Data Scientist with an M.S. in Analytics and experience building machine learning models, automating data pipelines, and delivering actionable business insights. Skilled in Python, SQL, and R, with experience in statistical modeling, LightGBM, and AI-powered document extraction. A proactive problem-solver who combines analytical thinking with leadership, cross-functional collaboration, and a focus on measurable impact.

Programming: Python, R, SQL, VBA
Software: Excel, Tableau, SPSS, Arena Simulation Software, Salesforce, AWS, Google Cloud Platform, AI Tools

EDUCATION

M.S. in Analytics May 2025
Institute for Advanced Analytics | NC State University | Raleigh, NC

B.S. in Statistics and Analytics; B.A. in Psychology; Minor in Neuroscience *with Highest Distinction* May 2023
University of North Carolina | Chapel Hill, NC

WORK EXPERIENCE

Ezenics, Inc. | Data Scientist | Remote (Chicago, IL) July 2025—Current

- Redesigned and automated client savings reporting workflows using **Python** and **SQL**, developing client-specific utility bill extraction and calculation pipelines that achieved a 0.96 F1 score in AI-powered extraction and reduced processing time for a major client from approximately one week to 4–5 hours.
- Independently developed and automated a **Python** pipeline to train 200 location-specific LightGBM quantile regression models, reducing model development time from several weeks to minutes and evaluating performance using MAE, MAPE, and custom peak-focused metrics.
- Enabled independent ownership of an automated reporting workflow by training a colleague, creating technical documentation and instructional videos, and providing support throughout the project handoff.
- Develop and evaluate machine learning models to forecast next-day peak electricity demand at the individual-building level, engineering features from weather forecasts and historical demand data and benchmarking LightGBM, histogram-based gradient boosting, and Lasso regression; achieved 3–4% average MAPE across 5 pilot locations.
- Present monthly client savings analyses quantifying millions of dollars in savings to approximately 20 client stakeholders, translating month-over-month and year-over-year trends into actionable business insights.

ADRMortgage.com | Data Analyst Contract | Raleigh, NC September 2023—January 2024

- Identified significant decreasing trends in groundwater reserves across the U.S. using Theil-Sen regression, informing the assessment of cities likely to impose building restrictions due to groundwater shortages.
- Collected, cleaned, and analyzed data in **R** from multiple databases, examining relationships between groundwater decline and factors such as precipitation, irrigation use, and population size.

Mercalis | Data and Reporting Analyst Intern | Morrisville, NC May 2022—August 2022

- Designed and implemented automated dashboards in **Salesforce** to present data on pharmaceutical products and health insurance cases, replacing manual **Excel** reports and improving data accessibility and operational efficiency.
- Applied row and summary-level formulas to overcome date-time limitations in **Salesforce** reporting, enabling data-driven analysis of monthly averages for pharmaceutical products.
- Optimized daily reporting workflow by researching, learning, and building **VBA** scripts to automate daily reports, cutting report preparation time by 80% and significantly reducing the level of effort (LOE) for routine tasks.
- Managed delivery of key metrics (KPIs) to internal stakeholders by organizing data exported from **Salesforce** into **Excel**, ensuring timely and accurate reporting over variable frequencies (monthly, weekly, daily).

GRADUATE PROJECT

Elevate | Team Lead September 2024—May 2025

- Led a team of 5 to investigate the decline in Direct Mail (DM) offer response rates by identifying missing customer populations, utilizing **SQL**, **Python**, and **R** for data extraction, manipulation, and analysis.
- Applied clustering for unsupervised variable reduction to credit data containing 10,000+ variables and 10 million customer records, reducing dataset dimensionality by 95%.

- Developed and evaluated linear and nonlinear machine learning models in **Python** to identify potential new responders, contributing to a projected 30% increase in response rates.
- Analyzed historical DM model predictions to identify key features associated with customer response behavior.

RESEARCH EXPERIENCE

UNC Emotions and Social Interactions Laboratory | Research Assistant | Chapel Hill, NC July 2021—May 2022

- Conducted participant sessions to collect biometrics and survey responses, utilizing interpersonal skills to ensure effective communication and analytical acumen to conduct detailed observations in **Excel**.

Games & Society Lab at Uppsala University | Research Assistant | Uppsala, Sweden May 2021—July 2021

- Co-led a 6-week research project investigating whether children overlook subtractive solutions across 2 experimental tasks, resulting in acknowledgment in a published paper.
- Recruited 20+ participants and conducted statistical analysis with **Excel** and **SPSS**, yielding significant results.